SL-R3528BRSIRPIRC-E3CH-AW

3.5x2.8mm, Multi-color LEDSurface
Mount PLCC-6 LEDs

SEAM Semiconductors

Technical Data Sheet

Features

- PLCC-6 package.
- White package.
- Optical indicator.
- Ideal for backlight and light pipe application.
- Wide viewing angle.
- Suitable for automatic placement equipment.
- Available on tape and reel (8mm Tape).
- The product itself will remain within RoHS compliant Version.

Descriptions

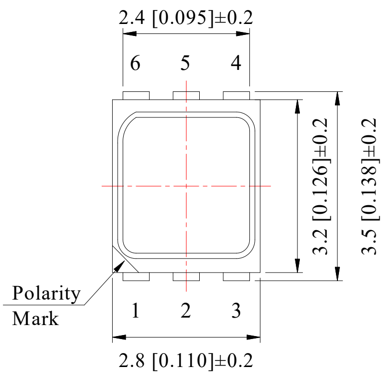
The R3528B series is available in soft red, orange, yellow, green, blue and white. Due to the package design, the LED has wide viewing angle and optimized light coupling by inter reflector. This feature makes the SMT TOP LED ideal for light pipe application. The low current requirement makes this device ideal for portable equipment or any other application where power is at a premium.

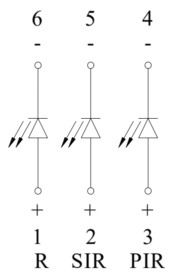
Applications

- Backlight in dashboards and switches.
- Telecommunication: Indicator and backlight in telephone and fax.
- Indicator and backlight for audio and video equipment.
- Indicator and backlight in office and family equipment.
- Flat backlight for LCD's, switches and symbols.
- Light pipe application.
- General use.

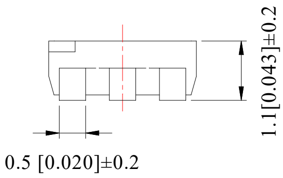
Part No.	Emitting Color		Lens Color
SL-R3528BRSIRPIRC-E3CH-AW	R	Red	Water Clear
	SIR	Infrared	
	PIR	Infrared	

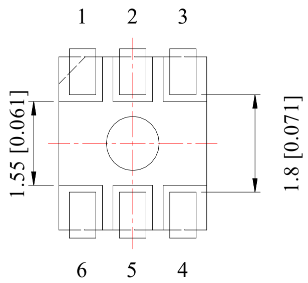
Package Dimension



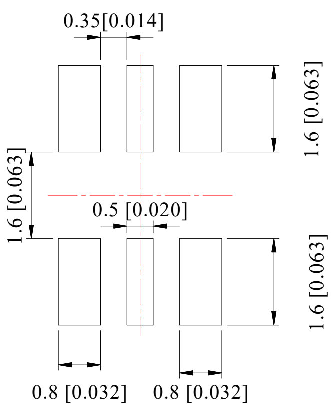


Polarity





Recommended Soldering Pad Dimensions



Notes:
1. All dimensions are in millimeters (inches).
2. Tolerance is ± 0.25 mm (.010″) unless otherwise noted.

Absolute Maximum Ratings at Ta=25℃

Parameters		Symbol	Max.	Unit
Power Dissipation	R	PD	60	mW
	SIR		60	
	PIR		60	
Peak Forward Current (Per Chip) [a]		IFP	100	mA
Forward Current (Per Chip) [b]	R	IF	25	mA
	SIR		25	
	PIR		25	
Reverse Voltage (Per Chip)		VR	5	V
Electrostatic Discharge (HBM)	R	ESD	2000	V
	SIR		2000	
	PIR		2000	
Operating Temperature Range		Topr	-40℃ to +85℃	
Storage Temperature Range		Tstg	-40℃ to +85℃	
Lead Soldering Temperature		Tsld	260℃ for 5 Seconds	

Notes:
a. Duty Factor = 10%, Frequency = 1 kHz.
b. Derate linearly as shown in derating curve.

Electrical Optical Characteristics at Ta=25℃

Parameters	Symbol	Emitting Color	Min.	Typ.	Max.	Unit	Test Condition
Radiant Power[a]	Φ_e	R	12	18	---	mW	IF=20mA*3
		SIR	10	15	---		
		PIR	5	10	---		
Viewing Angle	$2\theta_{1/2}$	R	---	120	---	Deg	IF=20mA*3
		SIR	---	120	---		
		PIR	---	120	---		
Peak Emission Wavelength[b]	λ_p	SIR	---	830	---	nm	IF=20mA*3
		PIR	---	1070	---		
Dominant Wavelength[b]	λ_d	R	630	633	635	nm	IF=20mA*3
Spectral Line Half-Width	$\Delta\lambda$	R	---	20	---	nm	IF=20mA*3
		SIR	---	22	---		
		PIR	---	20	---		
Forward Voltage[c]	VF	R	1.60	2.00	2.40	V	IF=20mA*3
		SIR	1.20	1.40	1.60		
		PIR	1.10	1.30	1.60		
Reverse Current	IR	R SIR PIR	---	---	10	μA	V_R=5V

Notes:

a. Radiant Power measurement tolerance: ±10%.

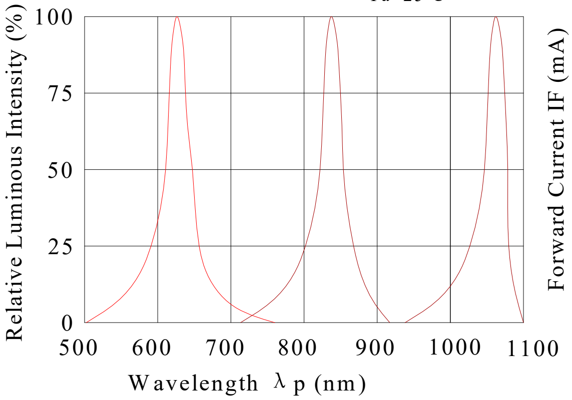
b. Wavelength measurement tolerance: ±1nm

c. Forward voltage measurement tolerance: ±0.1V

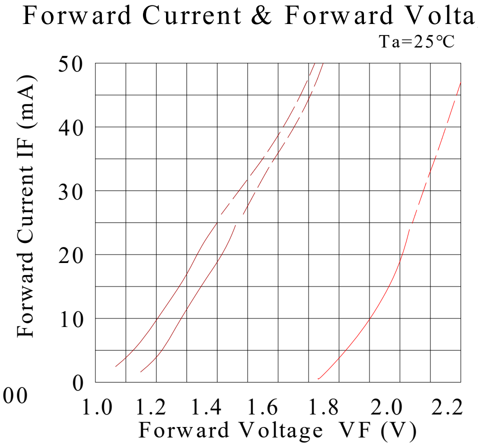
Technical Data Sheet

Typical Electrical / Optical Characteristics Curves
(25℃ Ambient Temperature Unless Otherwise Noted)

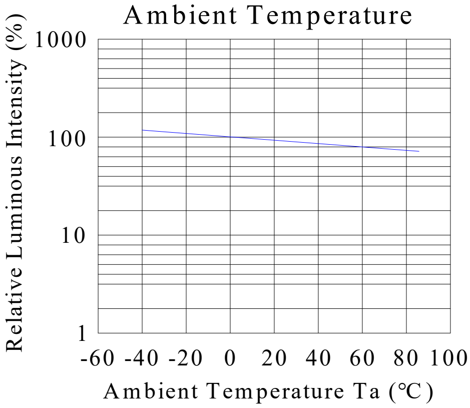
Spectrum Distribution



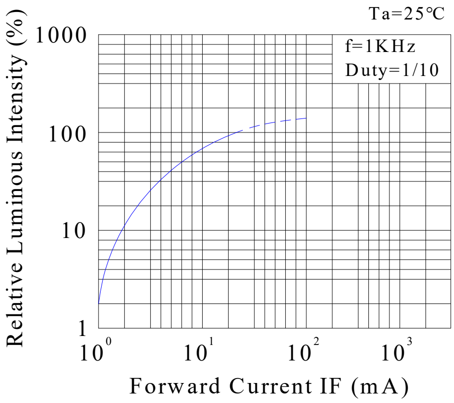
Forward Current & Forward Voltage



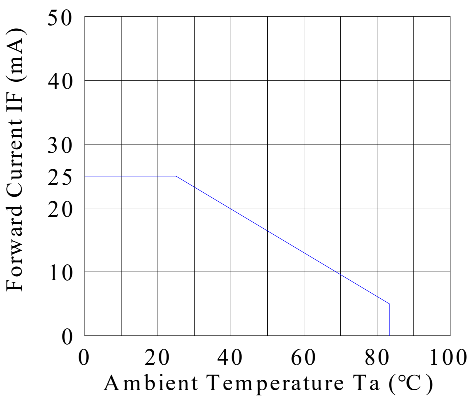
Luminous Intensity & Ambient Temperature



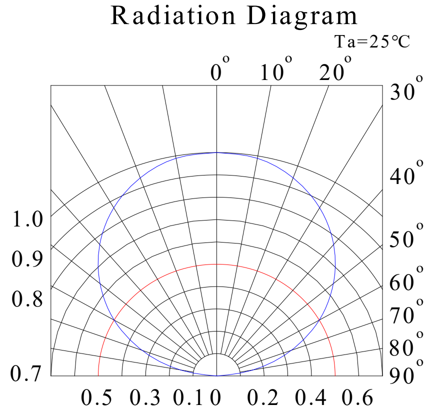
Luminous Intensity & Forward Current



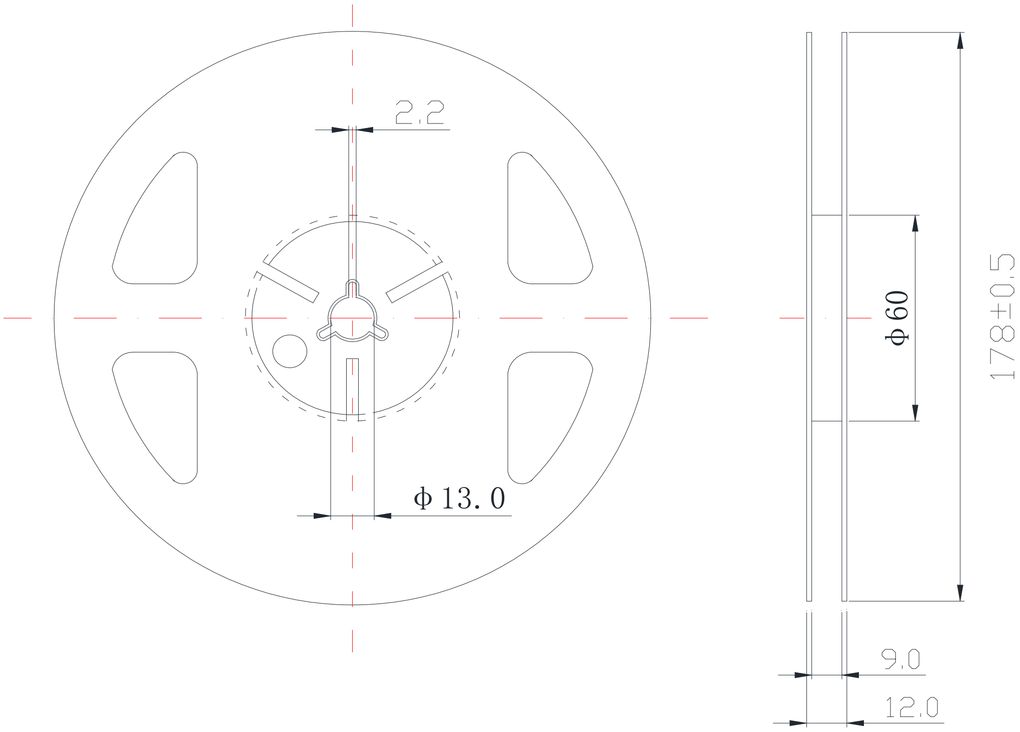
Forward Current Derating Curve



Radiation Diagram

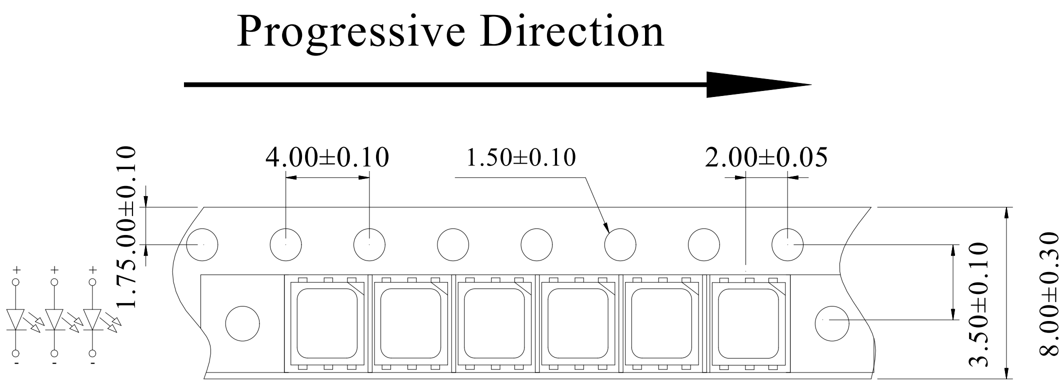


Reel Dimensions



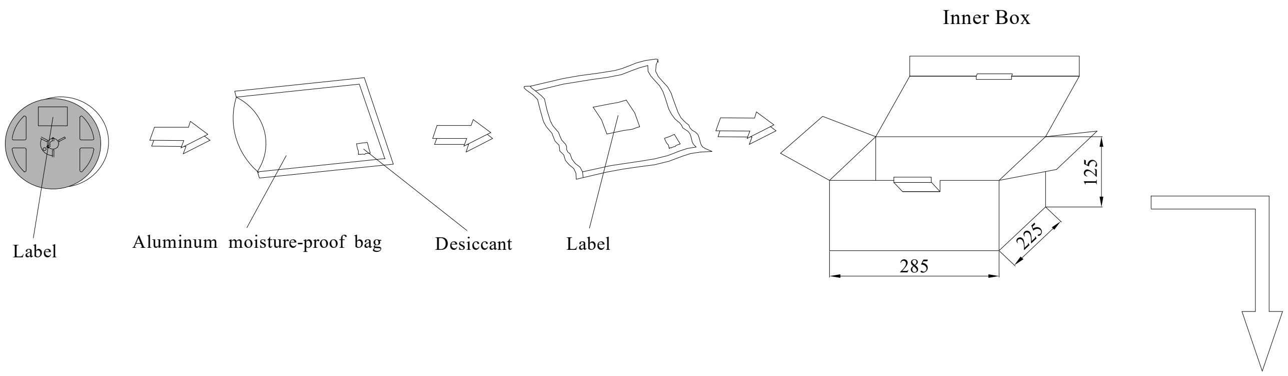
Carrier Tape Dimensions

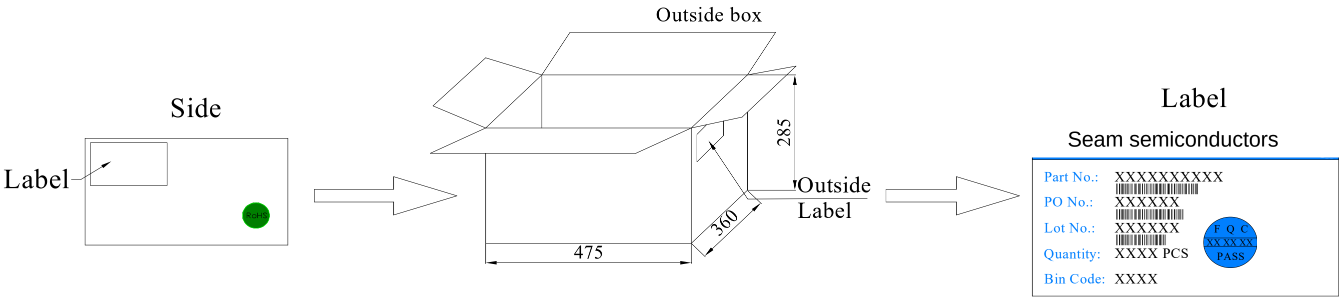
Loaded quantity 3000 pcs per reel.



Packing & Label Specifications

Moisture Resistant Packaging

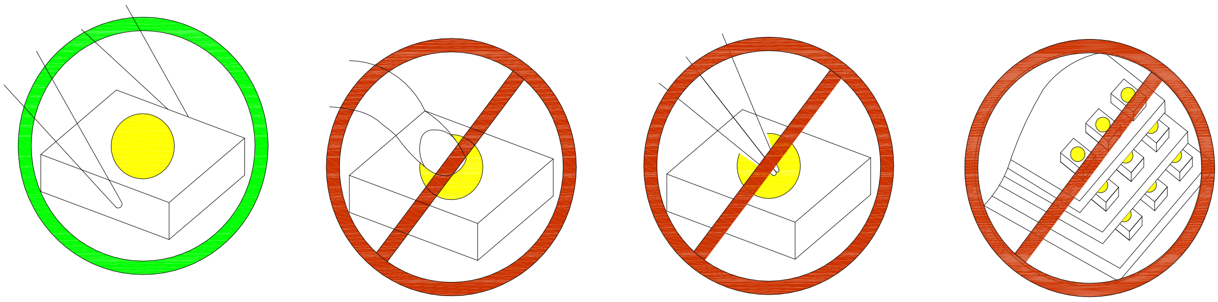




CAUTIONS

1. Handling Precautions:

1.1. Handle the component along the side surfaces by using forceps or appropriate tools.
1.2. Do not directly touch or handle the silicone lens surface. It may damage the internal circuitry.
1.3. Do not stack together assembled PCBs containing exposed LEDs. Impact may scratch the silicone lens or damage the internal circuitry.



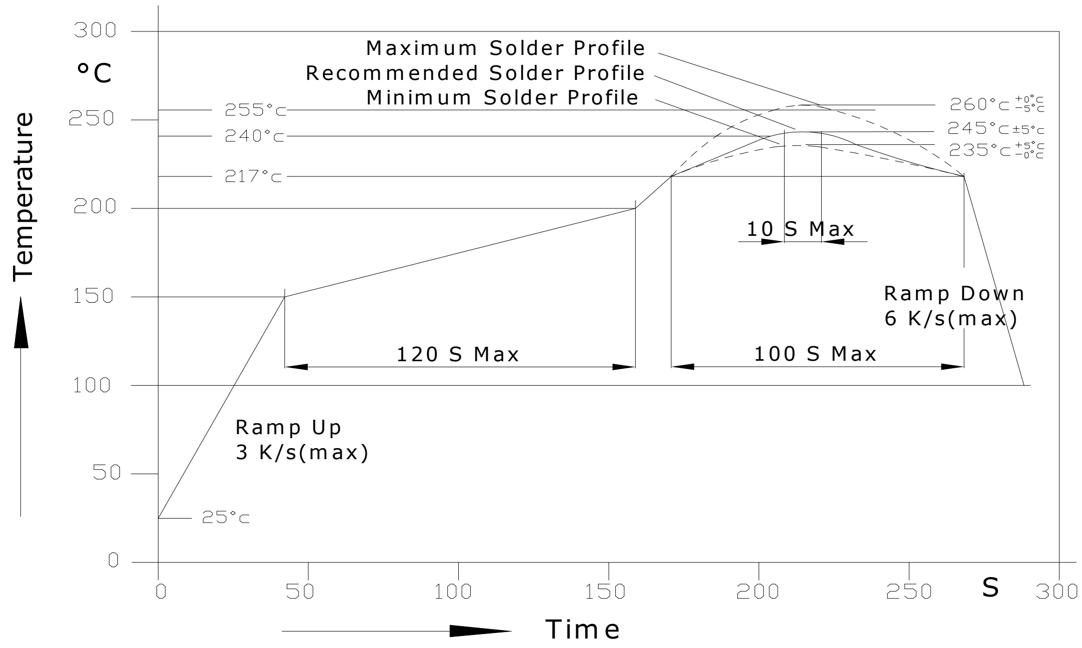
Compare to epoxy encapsulant that is hard and brittle, silicone is softer and flexible. Although its characteristic significantly reduces thermal stress, it is more susceptible to damage by external mechanical force. As a result, special handling precautions need to be observed during assembly using silicone encapsulated LED products. Failure to comply might lead to damage and premature failure of the LED.

2. Storage

2.1. Do not open moisture proof bag before the products are ready to use.
2.2. Before opening the package, the LEDs should be kept at 30°C or less and 60%RH or less.
2.3. The LEDs should be used within a year.
2.4. After opening the package, the LEDs should be kept at 30°C or less and 60%RH or less.
2.5. The LEDs should be used within 24 hours after opening the package.
2.6. If the moisture adsorbent material has fabled away or the LEDs have exceeded the storage time, baking treatment should be performed using the following conditions. Baking treatment: 65±5°C for 24 hours.

3. **Soldering Condition**

3.1. Pb-free solder temperature profile



3.2. Reflow soldering should not be done more than two times.
3.3. When soldering, do not put stress on the LEDs during heating.
3.4. After soldering, do not warp the circuit board.
3.5. Recommended soldering conditions:

Reflow soldering		Soldering iron	
Pre-heat Pre-heat time Peak temperature Soldering time	150~200°C 120 sec. Max. 250°C Max. 10 sec. Max.(Max. two times)	Temperature Soldering time	300°C Max. 3 sec. Max. (one time only)

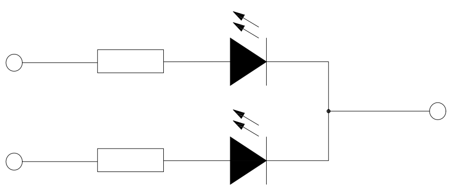
3.6. Because different board designs use different number and types of devices, solder pastes, reflow ovens, and circuit boards, no single temperature profile works for all possible combinations.

However, you can successfully mount your packages to the PCB by following the proper guidelines and PCB-specific characterization.

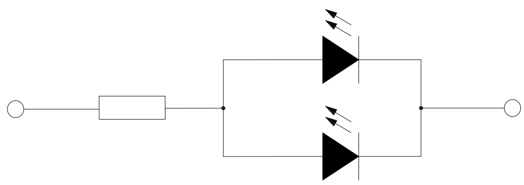
4. **Drive Method**

4.1. An LED is a current-operated device. In order to ensure intensity uniformity on multiple LEDs connected in parallel in an application, it is recommended that a current limiting resistor be incorporated in the drive circuit, in series with each LED as shown in Circuit A below.

Circuit model A
LED



Circuit model B
LED



a. Recommended circuit.

b. The brightness of each LED might appear different due to the differences in the I-V characteristics of those LEDs.

5. ESD (Electrostatic Discharge):

Static Electricity or power surge will damage the LED. Suggestions to prevent ESD damage:

- Use of a conductive wrist band or anti-electrostatic glove when handling these LEDs.
- All devices, equipment, and machinery must be properly grounded.
- Work tables, storage racks, etc. should be properly grounded.
- Use ion blower to neutralize the static charge which might have built up on surface of the LED's plastic lens as a result of friction between LEDs during storage and handling.

ESD-damaged LEDs will exhibit abnormal characteristics such as high reverse leakage current, low forward voltage, or "no lightup" at low currents. To verify for ESD damage, check for "lightup" and Vf of the suspect LEDs at low currents. The Vf of "good" LEDs should be >2.0V@0.1mA for InGaN product and >1.4V@0.1mA for AlInGaP product.

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